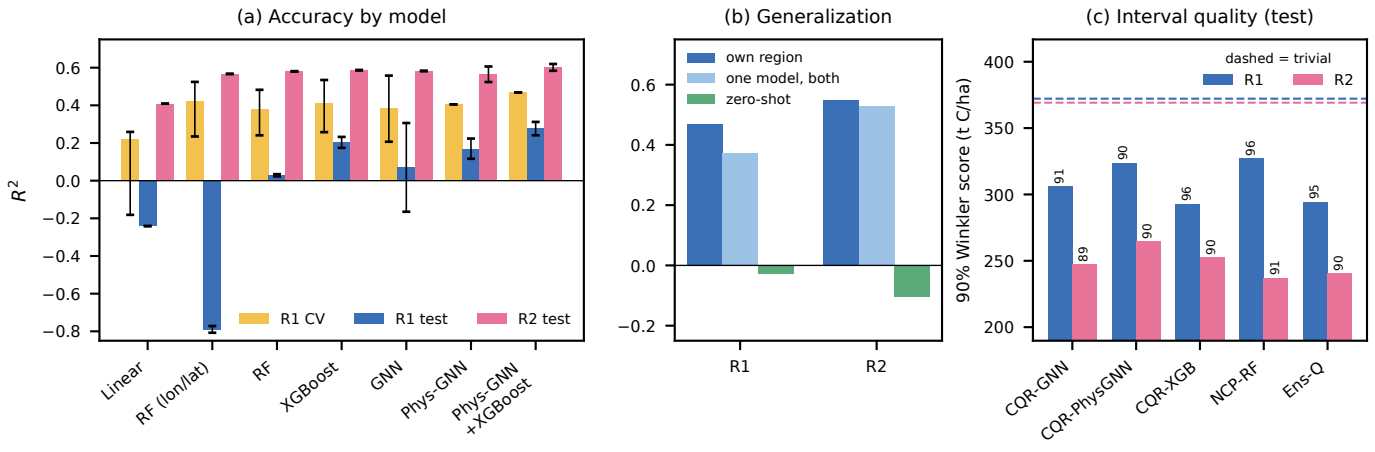
(a) Accuracy by model
R^2
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
R1 CV
R1 test
R2 test
Linear
RF (lon/lat)
RF
XGBoost
GNN
Phys-GNN
Phys-GNN +XGBoost
(b) Generalization
own region
one model, both
zero-shot
0.6
0.4
0.2
0.0
−0.2
R1
R2
(c) Interval quality (test)
90% Winkler score (t C/ha)
dashed = trivial
R1
R2
400
350
300
250
200
91
89
90
90
96
90
96
91
95
90
CQR-GNN
CQR-PhysGNN
CQR-XGB
NCP-RF
Ens-Q